



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000
WEEK 4.02

TUTOR GUIDE

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The cases and materials in this guide were reviewed by;

- ☐ Members of the University of Notre Dame critical care working group, a working group of the clinical year's curriculum committee [CYC]. Last reviewed Nov 2010.

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RELEVANT PRIOR LEARNING

MED1000

- 1.02 Principles of Fluid Replacement (L)
- 1.22 Management of Fluid balance (L)

MED2000

- 2.14 Assessment and management of GIT haemorrhage (CCS)
- 2.16 Fluid and Electrolyte Disturbances
- 2.19 Pharmacology of General Anaesthetic drugs (L)
- 2.27 Clinical problems in Acid Base (CCS)
- 2.27 Management of Fluid Balance (CCS)

WEEK 02_ ANAESTHESIA		
Yr	Wk	LEARNING OBJECTIVES
4	2	• Demonstrate an understanding of fluid and electrolyte physiology by being able to calculate and chart the appropriate fluid replacement & maintenance therapy for a patient presenting with an acute dehydrating illness. • Know the principles of when a patient is required to have a group and hold or blood cross matched in preparation for the operating theatre. • Know the principles of collection of a blood sample for group & hold or cross match. • Know the potential risks of blood transfusion and be able to explain these to a patient.
4	2	• Review the immunology of anaphylaxis and be able to correlate the clinical manifestations with the underlying pathophysiology • Know the key clinical parameters in the assessment of an anaesthetised patient experiencing an anaphylactic reaction. • Outline the management of anaphylaxis in the context of the anaesthetised patient. • Outline the assessment and management plan for a patient following an anaphylactic reaction during surgery under a general anaesthetic.
4	2	• Outline the key clinical features of malignant hyperthermia and describe the underlying pathophysiology. • Outline the principles of management of malignant hyperthermia. • Detail the management plan for a patient who has experienced malignant hyperthermia, for subsequent anaesthetics.
4	2	• Know the minimum standards of monitoring during anaesthesia and explain why these monitoring parameters are required. • Outline the options for non-invasive and invasive monitoring under anaesthesia and demonstrate an understanding of the use of different modes of monitoring by being able to outline a monitoring plan for a patient.

CASE ONE

Short case number: 4_02_01

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Emergency Department

Topic: Fluid and Electrolyte balance.

CASE



Timothy Mooney, a previously well 20 year old develops a small bowel obstruction due to a band adhesion and has severe vomiting for several days before presentation. His past medical history includes an Appendicectomy at age 10 years,

He is scheduled for laparotomy as he shows signs of bowel ischaemia.

QUESTIONS

1. You are the intern in the Emergency department; Timothy requires fluid resuscitation prior to theatre. How is the amount of fluid a patient requires calculated?
2. How would you assess Timothy's fluid deficit?
3. Outline how maintenance fluid requirements are calculated?
4. How is the type of fluid determined?
5. The surgical registrar asks you to organise a 'blood group & hold' why would this be necessary? Timothy asks you if it is likely that he will need a blood transfusion
6. Outline the clinical situations where a blood transfusion might be necessary for Timothy?
7. Timothy appears concerned about the risks of having a blood transfusion, particularly contracting a serious infection. What would you explain to him about the risks and complications of blood transfusion?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.
- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Sydney: Elsevier.

Electronic Resources (Available on Blackboard):

- Obtaining informed consent for blood and blood products (2008). ARCBS: www.transfusion.com.au

ANSWERS

How is the amount of fluid a patient requires calculated?

In general, fluid requirement is i. **Preexisting deficit** (see 2.), PLUS
ii. **Ongoing losses** PLUS
iii. **Maintenance** requirements

How would you assess Timothy's fluid deficit?

Assessment of deficit. Two things to consider – Total Body Water and Blood Volume.

Fluid compartments. For 70 kg young adult male.

TBW 60% 42litres

ECF 20% 14litres

Blood Vol 70ml/kg approx 5litres. Remember **Blood is only around 60% water** (depends on haematocrit)

For female and the elderly, less H₂O, more fat, less Blood Volume say 60mls/kg

Infants higher, say 70% H₂O and 85mls/kg

Water Deficit.

Mild dehydration. Est **3%** or 30ml/kg. Thirst; tachycardia; concentrated urine

Moderate . Est **6% or** 60ml/kg. Tachycardia, reduced (slow) skin turgor; slow capillary refill; oliguria; lethargy; postural hypotension; dry mucosae (beware mouth-breathing); depressed fontanelle

Severe Est **10%** or 100ml/kg. Obtunded; “shut-down”; anuric; hypotensive, tachycardic; “tenting” skin turgor; parched mucosae; absent tears;

Blood loss. 10%. You give this at the blood bank. Effects depend on other comorbidities and medications.

20%. Tachycardia, postural hypotension, oliguria. Thirst.

40% Shock. Anuria Obtundation

Outline how maintenance fluid requirements are calculated?

Maintenance. Water. **30-40ml/kg/day.** Less if elderly, hypothermic, paralysed, hypothyroid. More if febrile, active, burns (special case), children (see below)

Electrolytes. Na and K. **1-2mEq/kg/day.** Sick people with increased ADH are prone to develop

Hyponatraemia due to water retention. Avoid large volumes of hypotonic fluids, but conversely large volumes of isotonic fluids tend to cause oedema.

Children. General rule of thumb. 100ml/kg/day for first 10kg; 50 ml/kg/day for next 10 kg and 20ml/kg/day after that. Eg 30kg 9yo : 10kgx100mls + 10kgx50mls + 10 kgX 20 mls = 1,700 mls/day.

How is the type of fluid determined?

What Fluid? General rule : Give what is being lost. Care with diarrhoea!

- **Gastric fluid** : eg vomiting, pyloric stenosis. NaCl plus HCl plus KCL. Replace with equal volume of isotonic NaCl (not the time to go into metabolic alkalosis, paradoxical aciduria, role of hypokalaemia in perpetuation of met alkalosis, etc)
- **Small bowel** distal to Ligament of Trietz. Eg fistula. Fluid rich in HCO₃ from pancreas. Replace with Hartmanns Solution.
- **Large Bowel**. Eg diarrhoea .More akin to Hypotonic saline with added Potassium eg 4%Dextrose with 0.18%Saline
- **Polyuria**. Often this occurs in the setting of recovery from severe illness and is simply clearing a salt and water load due to resuscitation. If pathological(eg Diabetes insipidus, or polyuric phase of acute renal failure) easy to measure electrolyte content of the urine and replace with appropriate fluids.
- **Sweat**. Aldosterone affects sodium content but in unacclimatised, high Na so replace with isotonic saline initially.
- **Burns**. Various protocols. Lose enormous amount of water, electrolytes and albumin.

Outline the clinical situations where a blood transfusion might be necessary for Timothy?

Transfusion Triggers depend on patient's co-morbidities, and expected blood loss. In general, a healthy adult will tolerate a Hb of 70-80g/l provided there is no ongoing bleeding. A patient with impairment of another aspect of Oxygen delivery (oxygenation or cardiac output) will require a higher Hb. Patients with coronary artery disease tolerate anaemia less well than other patients. The severely traumatised patient requires empirical blood and blood product transfusion, often prior to the availability of formal testing.

What would you explain to him about the risks and complications of blood transfusion?

Complications of Blood Transfusion are i. Those that can occur with ANY unit eg Acute Haemolytic reaction (ABO mismatch); alloimmunisation (rH); immune modulation; disease transmission (HCV, HIV, bacteria, others), TRALI, Delayed (immune) haemolytic reaction; and those specific to massive transfusion (greater than one blood volume in a 24 hour period). These include platelet and coagulation factor depletion, especially the so-called labile factors 2,7,9 and 10; hypothermia; hyperkalaemia; hypocalcaemia rarely (citrate toxicity); Hyperbilirubinaemia down the track.

CASE TWO

Short case number: 4_02_02

Category: Clinical skills

Discipline: Anaesthesia

Setting: Operating Theatre

Topic: Anaesthetic emergencies_anaphylaxis

CASE



You have just anaesthetized Peter, an otherwise healthy 25 year old male, for removal of four wisdom teeth. He has been induced and intubated with a nasal endotracheal tube.

The surgeon is prepping and draping when you notice a distinct rise in heart rate. The NIBP is unable to record a blood pressure. The end-tidal carbon dioxide measurement has dropped and you notice a growing redness of the exposed skin around his mouth.

The drapes are removed and reveal marked erythema of his face, neck arms and torso.

QUESTIONS

1. Your working diagnosis is anaphylaxis. What is the pathophysiology of this condition? What are its clinical manifestations, and which of these are likely to be detected under anaesthesia?
2. What is your approach to management?
3. What are the common triggers of anaphylaxis in the context of a general anaesthetic and operating theatre environment?
4. What follow-up should be arranged for Peter following an episode of suspected anaphylaxis?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.
- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Sydney: Elsevier.

ANSWERS

What are the clinical features of anaphylaxis, with particular reference to the anaesthetised patient. What features of this case make anaphylaxis a likely diagnosis? What agents are most commonly implicated?

Acute reaction to a chemical or chemical complex that is recognised as hostile. Variable symptoms and signs but classically bronchospasm; profound hypotension due to vasodilatation; diffuse erythematous flush; urticaria; angioneurotic oedema;; GIT symptoms of nausea, abdo pain, diarrhoea. Onset within 5 minutes for parenteral agents but may be delayed for hours. Duration varies from transient to days. Common agents outside hospital include bee stings, peanuts;; drugs include iodinated contrast media (probably leading cause of death); penicillin; latex. Neuromuscular blockers , colloid intravenous fluids. Incidence difficult to quantify due to reporting being voluntary.

Describe the pathophysiology of anaphylaxis and anaphylactoid reactions

Anaphylaxis classically with prior exposure leading to sensitisation by IgE binding to surface of mast cells and basophils. Subsequent exposure leads to mast cell degranulation and release of histamine, PAF, interleukins, ECF-A and other mediators. Overall effect is vasodilatation, glandular secretion, increased capillary permeability. The term “anaphylactoid” is best not used for the **identical syndrome** caused by non-immune mast cell degranulation. Direct histamine-release by drugs indistinguishable clinically and treatment identical. Probably best to use terms like “immune” and “non-immune” anaphylaxis. Pathophysiology of the hypotension a little unclear how much contribution from myocardial depression, though clinically and on Echo, see empty, normally-contracting heart.

Treatment . No RCT's due to unexpected onset, rapid course, and (usually) rapid response to treatment. Oxygen. Intubation if airway oedema. Ventilation if bronchospasm severe. Adrenaline. In the community, give imi eg 0.3 – 1mg. In hospital or if muscle blood flow thought to be compromised (eg CPR) give ivi as 3 – 5 ml of 1/10,000 solution (Minijet). Further doses needed in 1/3 and infusion for some minutes or hours in 10%. Adrenaline universally recommended – beneficial effect on vascular tone as well as inhibiting further mediator release. Refractory hypotension has been managed with noradrenaline and /or vasopressin . Intravenous fluids preferably as colloids. Corticosteroids have no proven benefit and should be reserved for refractory bronchospasm (nor is there evidence of harm, and so usually end up being given anyway!); H1 and H2 blockers are often given .

Outline the management including testing after recovery

Follow up. Mast cell tryptase . Elevated from 1 – 4 hours after reaction. Can be used post-mortem. Highly sensitive and specific. Testing by intradermal skin-prick >4 weeks after reaction. If positive, a “medic-alert” bracelet should be worn.

CASE THREE

Short case number: 4_02_03

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Operating Theatre

Topic: Anaesthetic Emergencies_malignant hyperthermia

CASE



Chandler Ackroyd a 16 year old boy has a lengthy operation for a post-nasal space angiofibroma. In the recovery room a large neck haematoma develops and he is returned to the operating room for exploration.

He is given Suxamethonium to facilitate reintubation and the wound is reopened. Over the next ½ hour his urine becomes dark reddish-black, CO₂ rises to 75mmHg and he

develops multifocal VEBs on the ECG.

His temperature is measured - it is 39.5⁰C. Malignant Hyperthermia is diagnosed and appropriate treatment instituted.

He survives and returns to OR one week later for a planned second-stage procedure.

QUESTIONS

1. What is malignant hyperthermia (MH)? Outline the pathophysiology of MH, including changes at a cellular level?
2. What diseases are associated with MH susceptibility?
3. Describe the clinical presentation and describe the principles of management.
4. Chandler's parents are very concerned that this could happen again with his next Anaesthetic. Explain the how the possible malignant hyperthermia syndrome patient can be tested and managed for subsequent anaesthetics.

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.
- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Sydney: Elsevier.

ANSWERS

What is malignant hyperthermia (MH)? Outline the pathophysiology of MH, including changes at a cellular level?

MH is a rare pharmacogenetic disorder manifest when a susceptible individual (MHS) is exposed to a triggering agent. Characterised by hypermetabolic state, skeletal muscle rigidity, with pyrexia, respiratory and metabolic acidosis, and hyperkalaemia, arrhythmias, and rapid death in some cases. Incidence approx 1/60,000 with suxamethonium; 1/250,000 with volatiles. Children more common – 50% of cases <15 years old. Pathogenesis involves sarcolemmal defect in skeletal muscle only, such that Ryanodine receptor when triggered leads to massive release of Ca from sarcoplasmic reticulum. ATP-dependent membrane pumps then attempt to return this Ca to the SR, leading to sustained glycolytic and aerobic metabolism. Return of Ca is incomplete, leading to sustained contraction and rigidity. Eventually oxidation and phosphorylation are uncoupled, anaerobic metabolism occurs, and severe mixed acidosis occurs.

What diseases are associated with MH susceptibility?

Genetic component. Central core disease and mitochondrial myopathies good evidence. Poor evidence for DMD and other myopathies, and for patients who have had NMS or severe heat stroke. Triggering agents include all Volatile anaesthetic agents; suxamethonium; Ca-channel blockers.

Describe the clinical presentation and describe the principles of management

Clinical presentation initially rise in ET CO₂ and rapid exhaustion of soda-lime; tachycardia. ?Masseter spasm after Suxamethonium?. Leading on to hyperthermia ; hyperkalaemia ; acidosis; hypoxaemia; later dark urine (myoglobin); VEB's; CK rise; death.

Management requires several people. First step is to suspect it. Then stop all triggers. Stop the surgery if possible. Ventilate with high minute volumes from a non-triggering source. Give Dantrolene 2.5mg/kg initial dose and repeat every 10-15 minutes until biochemical changes are reversed. Dantrolene is the only specific treatment and all hospitals should have access to an initial dose. Treat arrhythmias and hyperkalaemia in conventional ways but avoid Ca-channel blockers.. Cooling with tepid sponging, fanning, cold iv fluids; paralysis. Monitoring in ICU with management of rhabdomyolysis if necessary. Referral of patient and relatives for testing after recovery.

Explain the how the possible malignant hyperthermia syndrome patient can be tested and managed for subsequent anaesthetics.

Testing. Currently in flux between IVCT and DNA testing. IVCT very reliable – index case should certainly be confirmed this way. Open Quadriceps biopsy then test in bath with caffeine and/or Halothane and measure contraction developed. Too many variants of RYR1 gene at present for reliable DNA testing – up to 20% may be missed.

CASE FOUR

Short case number: 4_02_04

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Operating Theatre

Topic: Monitoring_standard & invasive

CASE



You have on your routine day-stay list for knee arthroscopy a morbidly-obese 40 year-old lady. She weighs 135kg and is 155cm tall. Unfortunately, there is no BP cuff that is both narrow enough for her quite short arms, and long enough to adequately appose the Velcro on the cuff; it keeps on “popping off” every time the automated BP machine inflates.

You ask the nurse to assemble the equipment you need to insert an intra-arterial line to allow you to measure the BP.

QUESTIONS

1. What is the minimum routine monitoring that is required for a patient undergoing general anaesthesia? What other (optional, or additional) monitoring tools are available?
2. What are the indications for invasive arterial pressure monitoring? Describe how this is performed.
3. What other invasive haemodynamic pressures can be monitored in the OR? What is Central venous pressure, what can affect it? Describe the waveform seen in a normal person.
4. What are some indications for, and complications of, central venous catheterisation?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.
- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Sydney: Elsevier.

Electronic Resources:

<http://www.anzca.edu.au/resources/professional-documents/ps18.html>

(ANZCA Recommendations on Monitoring during anaesthesia)

ANSWERS

1. What is the minimum routine monitoring that is required for a patient undergoing general anaesthesia? What other (optional, or additional) monitoring tools are available?

See ANZCA PS18. Basically, SpO₂, ETCO₂, ECG, NIBP, Oxygen, Volatile agent, Resp mechanics for all. Need available Temp, Neurological (BIS or Entropy)

2. What are the indications for invasive arterial pressure monitoring? Describe how this is performed.

A. Absolute – non-pulsatile arterial flow eg Cardiopulmonary Bypass; Left Ventricular Assist Device.

B. Relative i. Likely major BP swings or haemodynamic instability. Need for vasoactive drug infusion.

ii. Inability to measure reliably non-invasively eg the patient in this case, or Atrial fibrillation, or very slow heart rate

iii. Very prolonged surgery where there is a small risk of radial nerve damage from repeated cuff inflation.

iv. Surgery on both arms (can often use the leg unless obese)

v. Need for repeated blood sampling especially arterial blood gases.

Needs a catheter in an artery (usually radial or femoral); fluid-filled column (non-compliant tubing filled with saline); a transducer attached to a fast-flush device (transducer converts one form of energy to another, in this case mechanical to an electrical signal; the fast-flush device is a bag of saline in a pressure bag inflated to greater than the systolic pressure); and a means of processing and displaying the signal (a monitor). Requirements for accurate pressure reading are a short cannula; rigid tubing as short as possible between the patient and the transducer as free of air bubbles as possible; a transducer with as high a natural resonant frequency as possible (in practice, this is determined by the manufacturer); proper Zeroing ie the monitor needs to know what is the atmospheric pressure above which the BP is being measured, and the transducer must be at an appropriate level compared to the patient eg heart-height or circle of Willis height.

3. What other invasive haemodynamic pressures can be monitored in the OR? What is Central venous pressure, what can affect it? Describe the waveform seen in a normal person.

Other pressures measured include : Central Venous (CVP); Pulmonary Artery pressure and PA occlusion pressure ("Wedge pressure"); Intracranial (ICP); occasionally others like coronary sinus, internal carotid artery back-pressure.

CVP is the next most commonly-measured. It is the pressure in the right atrium or vena cava and is measured via a catheter inserted into a great vein (usually internal jugular or subclavian). It is a reflection of the volume in the circulation as a whole. It can be affected by

1. Intravascular volume (as 90% of volume is in the venous system)
2. Venous tone (ie venoconstriction will increase CVP)
3. Intra-thoracic pressure (so will be affected by positive-pressure ventilation)
4. Right heart function (Tricuspid valve and right ventricle)

Note that it is not affected by left-ventricular function and so is not a good indicator of LV loading conditions. There is no one "normal" CVP, rather a range that provides the best loading conditions in an individual. If BP is low, a trial of intravenous volume and re-assessment of CVP after that volume load is

often useful. General guide : if CVP <10, room for volume; CVP >12, probably “full” but may be worth a trial of volume (particularly with regard to those variables above that can affect CVP). Bottom line is not to treat the numbers, but to use the numbers as one more piece of information to enable an informed management decision.

CVP is often used, however, as a surrogate for LA pressure – there are many, many fallacies in this argument – however, using CVP as a guide to filling the heart after cardiac and major surgery is often very helpful.

CVP waveform. Normally a, c and v waves and x and y descents.. A wave from atrial contraction; c wave from TV closure; v wave from atrial filling in systole. X descent from movement of TV during RV systole; y descent from rapid emptying of RA when TV opens.

Abnormal patterns can be used to diagnose cardiac abnormalities eg V-wave (superimposed over c and v) with tricuspid regurgitation; large a wave with complete heart block.

4. What are some indications for, and complications of, central venous catheterisation?

Indications for central Venous line placement :

- a. No other venous access possible
- b. To monitor CVP eg major surgery or major blood or fluid shifts possible
- c. To allow infusion of irritant substances eg catecholamines; potassium solutions; vancomycin; TPN
- d. For long-term venous access
- e. For temporary or permanent haemodialysis access
- f. To allow pulmonary artery catheterisation
- g. To allow transvenous cardiac pacing

Complications of central lines.

- a. Damage to other structures – artery (carotid, subclavian,); pleura and lung; thoracic duct; vagus; recurrent laryngeal nerve; pericardial tamponade
- b. Arrhythmias either from guide-wire insertion or the catheter itself
- c. Infection. Minimise by use of: full aseptic insertion technique + antibiotic impregnated catheter + dedicated central-line team + dressing change only when indicated not routinely + access line only when necessary + prep with 2% Chlorhexidine in alcohol (not in operating rooms). Usually S aureus but can be anything also fungi.
- d. Thrombosis.
- e. Air embolus. Ensure lines only opened when patient supine.
- f. Catheter embolus (Not with modern catheter-over-guidewire systems)
- g. Misinterpretation of the data
- h. Patient harm due to inattention while inserting the line.